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Problem Set #1: Solutions and Empirical Analysis

Macroeconomic Measurement, National Accounts Decomposition, Goods Market Equilibrium, and Multipliers

1 Real GDP and the GDP Deflator (20 Points)

(1) Nominal and Real GDP Time Series (10 points)

Annual macroeconomic data are retrieved from the Census and Statistics Department (C&SD) of the Hong Kong Special Administrative Region (Table 310-31001: *Gross Domestic Product (GDP), implicit price deflator of GDP and per capita GDP*). We extract “GDP at current market prices” (nominal GDP, $Y_t^{\$}$) and “GDP in chained (2024) dollars” (real GDP, Y_t) for the historical observation window 1961–2025 (65 annual observations). Figure 1 plots both series, constructed analogously to Figure 2-1 in Blanchard (2021).

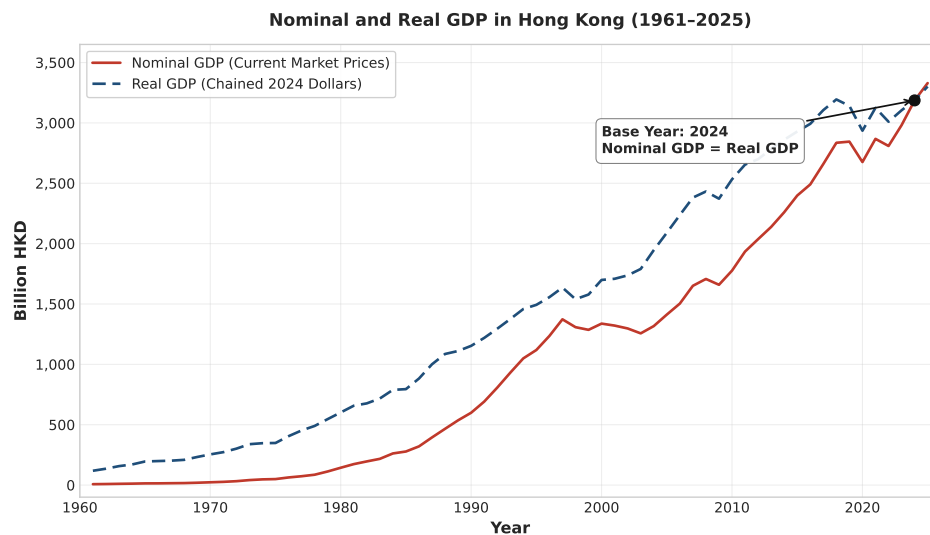


Figure 1: Nominal and Real GDP in Hong Kong (1961–2025)
Data Source: C&SD Table 310-31001. Real GDP is expressed in chained (2024) dollars with base year 2024.

Economic and Structural Interpretation:

- **Base Year Intersection:** The nominal and real GDP curves intersect precisely at the reference base year **2024**, where $\text{Nominal GDP}_{2024} = \text{Real GDP}_{2024} = \text{HK\$ } 3,186,526 \text{ million}$, by definition setting the implicit price deflator $P_{2024} \equiv 100$.
- **Pre-2024 Volume Dynamics ($P_t < 100$):** Prior to 2024, real GDP lies strictly above nominal GDP. Since real GDP measures production using constant base-year (2024) prices:

$$Y_t = \frac{Y_t^{\$}}{P_t/100}$$

Because historical price levels were substantially lower than the 2024 benchmark ($P_t < 100$), dividing by a price index less than 1 scales up the physical volume of production relative to its current-market-price valuation.

- **Post-2024 Trajectory ($P_t > 100$):** In 2025, nominal GDP (HK\$ 3,329,843 million) exceeds real GDP (HK\$ 3,301,474 million), reflecting cumulative positive price inflation relative to the benchmark year ($P_{2025} = 100.86$).

(2) Real GDP Growth Rates and Long-Term Average (5 points)

For each year $t \in [1962, 2025]$, the annual percentage growth rate of real GDP ($g_{Y,t}$) is given by:

$$g_{Y,t} = \frac{Y_t - Y_{t-1}}{Y_{t-1}} \times 100\% \quad (1)$$

Computing the arithmetic mean across all 64 continuous annual growth rates (identical to Excel's AVERAGE function):

$$\bar{g}_Y = \frac{1}{64} \sum_{t=1962}^{2025} g_{Y,t} = \mathbf{5.45\%} \quad (5.4468\%) \quad (2)$$

Macroeconomic Highlights: Hong Kong exhibited rapid double-digit growth during early export industrialization, peaking at +16.16% in 1976. Major cyclical troughs reflect severe external shocks: −5.88% during the 1998 Asian Financial Crisis and an all-time low of −6.54% during the 2020 pandemic shock.

(3) GDP Deflator and Inflation Rates: Discrete vs. Logarithmic (5 points)

The implicit GDP deflator (P_t , base year 2024 = 100) and the discrete percentage inflation rate (π_t) are defined as:

$$P_t = \frac{\text{Nominal GDP}_t}{\text{Real GDP}_t} \times 100, \quad \pi_t = \frac{P_t - P_{t-1}}{P_{t-1}} \times 100\% \quad (3)$$

Economists also frequently use the natural log-difference to approximate the instantaneous rate of inflation:

$$\pi_t^{\log} = \Delta \ln P_t \times 100\% = (\ln P_t - \ln P_{t-1}) \times 100\% \quad (4)$$

Both series are plotted concurrently in Figure 2.

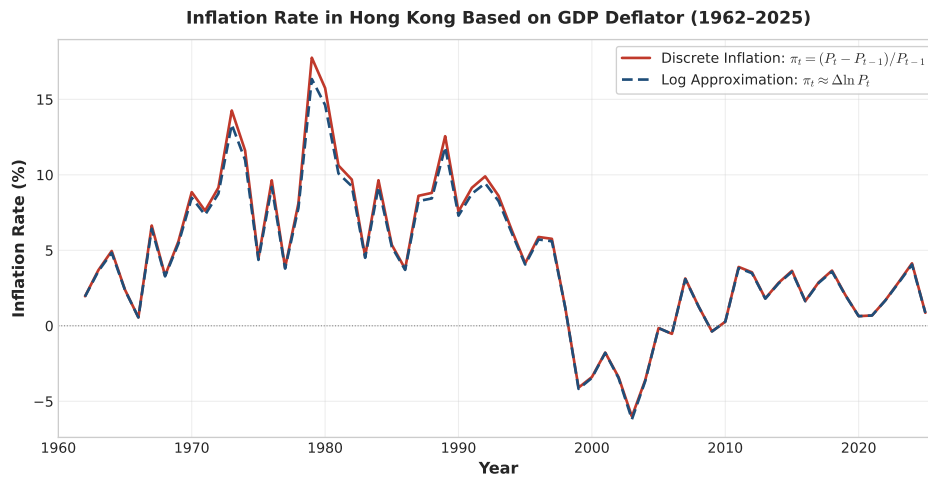


Figure 2: Comparison of Discrete and Logarithmic Inflation Rates (1962–2025)

Mathematical Foundation and Comparative Analysis:

- **Empirical Similarity:** The two series are almost indistinguishable over the 64-year sample, yielding a Pearson correlation coefficient $\rho = 0.9995$ and an average absolute discrepancy of merely 0.2035%.
- **Analytical Justification via Taylor Series:** Factoring the logarithmic difference:

$$\Delta \ln P_t = \ln \left(\frac{P_t}{P_{t-1}} \right) = \ln \left(1 + \frac{P_t - P_{t-1}}{P_{t-1}} \right) = \ln(1 + \pi_t) \quad (5)$$

Expanding $\ln(1+x)$ via a Maclaurin series for small inflation rates ($|\pi_t| \ll 1$):

$$\ln(1+\pi_t) = \pi_t - \frac{1}{2}\pi_t^2 + \frac{1}{3}\pi_t^3 - \cdots = \pi_t - \frac{1}{2}\pi_t^2 + \mathcal{O}(\pi_t^3) \quad (6)$$

The approximation error is dominated by $\frac{1}{2}\pi_t^2$. When inflation is moderate ($\pi_t \approx 0$), higher-order terms vanish, proving $\Delta \ln P_t \approx \pi_t$.

- **Regime Divergence:** Observable separation occurs strictly in high-inflation regimes. In 1979, discrete inflation peaked at +17.88% while log-inflation yielded +16.45% (a 1.43% gap, which closely matches the theoretical second-order penalty $\frac{1}{2}(0.1788)^2 \approx 1.60\%$). Conversely, during low-inflation regimes (e.g., 2015–2025), both measures align. The series also captures Hong Kong's protracted deflationary era from 1998 to 2004 ($\pi_t < 0$, troughing at -6.18% in 2003).

2 The Composition of HK GDP, 2023 (25 Points)

(4) Expenditure Components Accounting Table (10 points)

Using official data from Table 310-31002 (*Gross Domestic Product (GDP) by major expenditure component at current market prices*), the 2023 expenditure components are tabulated in Table 1. In modern national accounts, total gross investment (I) encompasses gross domestic fixed capital formation (I_{fixed}) and changes in inventories (ΔInv): $I = I_{\text{fixed}} + \Delta \text{Inv}$.

Table 1: Expenditure Components of Hong Kong GDP (2023, Current Market Prices)

Component	Millions of HKD, 2023	Percent of GDP (%)
Gross Domestic Product (Y)	2,981,210	100.00%
Consumption (C)	2,087,569	70.02%
Investment (I)	481,511	16.15%
<i>Gross domestic fixed capital formation</i> (I_{fixed})	505,510	16.96%
<i>Changes in inventories</i> (ΔInv)	−23,999	−0.81%
Government spending (G)	396,813	13.31%
Net Exports (NX)	15,317	0.51%
Exports of goods and services (X)	5,272,395	176.85%
Imports of goods and services (IM)	5,257,078	176.34%
Inventory investment (ΔInv)	−23,999	−0.81%

National Accounting Identity Balance Check:

$$\begin{aligned} C + I + G + NX &= 2,087,569 + 481,511 + 396,813 + 15,317 \\ &= 2,981,210 \equiv Y \quad (\text{Exact closure}) \end{aligned}$$

(5) Dominant Component of GDP (5 points)

Among the four primary expenditure components (C, I, G, NX), **Private Consumption Expenditure (C)** is the largest component, totaling **HK\$ 2,087,569 million** and accounting for **70.02%** of total GDP in 2023.

(6) Trade Openness Index and US Comparison (5 points)

From Table 310-31002, the trade and output statistics for Hong Kong in 2022 are:

$$\text{Exports of goods and services } (X_{2022}) = \text{HK\$ } 5,463,019 \text{ million} \quad (7)$$

$$\text{Imports of goods and services } (IM_{2022}) = \text{HK\$ } 5,348,126 \text{ million} \quad (8)$$

$$\text{Gross Domestic Product } (Y_{2022}) = \text{HK\$ } 2,808,922 \text{ million} \quad (9)$$

Evaluating the trade openness index:

$$\text{Openness}_{\text{HK},2022} = \frac{X_{2022} + IM_{2022}}{Y_{2022}} = \frac{5,463,019 + 5,348,126}{2,808,922} = \frac{10,811,145}{2,808,922} \approx \mathbf{3.85} \quad (\mathbf{384.89\%}) \quad (10)$$

Economic Comparison with the US Economy ($\approx 25\%$ in 2023): Hong Kong's openness index of **384.89%** is more than **15 times larger** than that of the United States. This divergence reflects fundamental structural differences:

1. **Global Entrepôt and Re-export Intermediation:** Hong Kong operates as an ultra-open international trading and logistics hub. Trade statistics reflect gross commercial turnover (X, IM), whereas GDP measures net domestic value-added (Y). Intermediate merchandise and re-exports moving through Hong Kong routinely exceed domestic economic value-added multiple times over.
2. **Continental Economy vs. Small Open Economy:** The US is a continental-scale, highly self-sufficient economy where interstate domestic trade forms the core economic engine; cross-border international trade naturally constitutes a modest fraction of aggregate GDP.

(7) Industry-Wise Decomposition (5 points)

From Table 310-34101 (*Gross Domestic Product (GDP) by major economic activity - percentage contribution to GDP*), the contributions of the five primary economic sectors in 2023 are:

1. Agriculture, fishing, mining and quarrying: 0.0%
2. Manufacturing: 1.0%
3. Electricity, gas and water supply, and waste management: 1.1%
4. Construction: 4.3%
5. **Services: 93.5%**

The overwhelmingly dominant industry is **Services**, accounting for **93.5%** of Hong Kong's GDP in 2023. Key drivers include Financing and Insurance (24.9%) alongside Import/Export, Wholesale and Retail Trades (17.5%).

3 The Employment Rate (10 Points)

(8) Derivation of the Employment Rate (10 points)

Let the macroeconomic labor aggregates be defined as:

- WAP: Working-Age Population
- L : Labor Force, where $L = E + U$ (E : Employed, U : Unemployed)

Following standard OECD definitions:

$$\text{Participation Rate (PR)} \equiv \frac{L}{\text{WAP}} = 0.80 \quad (11)$$

$$\text{Unemployment Rate (u)} \equiv \frac{U}{L} = \frac{L - E}{L} = 1 - \frac{E}{L} = 0.10 \implies \frac{E}{L} = 1 - u = 0.90 \quad (12)$$

$$\text{Employment Rate (ER)} \equiv \frac{E}{\text{WAP}} \quad (13)$$

Factoring the employment rate definition into the product of labor participation and employment intensity:

$$ER = \frac{E}{\text{WAP}} = \left(\frac{E}{L} \right) \times \left(\frac{L}{\text{WAP}} \right) = (1 - u) \times PR \quad (14)$$

Substituting the parameter values:

$$ER = (1 - 0.10) \times 0.80 = 0.90 \times 0.80 = \mathbf{0.72} \quad (\mathbf{72.0\%}) \quad (15)$$

The employment rate in this economy is **72%**. Note that $ER \neq 1 - u$ because the denominator of the unemployment rate is the active labor force (L), whereas the denominator of the employment rate is the total working-age population (WAP).

4 Goods Market Equilibrium and Multipliers (55 Points)

Consider the following goods market system for a closed economy (in billions of euros):

$$C = 480 + 0.5Y_D, \quad I = 110, \quad T = 70, \quad G = 250, \quad Y_D \equiv Y - T \quad (16)$$

(9) Goods Market Equilibrium Output (Y^*) (10 points)

In equilibrium, aggregate output equals aggregate demand ($Y = Z$):

$$Y = C + I + G = [480 + 0.5(Y - 70)] + 110 + 250 \quad (17)$$

$$Y = 480 + 0.5Y - 35 + 110 + 250 = 805 + 0.5Y \quad (18)$$

$$(1 - 0.5)Y = 805 \implies 0.5Y = 805 \implies Y^* = \frac{805}{0.5} = \mathbf{1,610} \text{ billion euros} \quad (19)$$

(10) Equilibrium Disposable Income (Y_D^*) (5 points)

$$Y_D^* = Y^* - T = 1,610 - 70 = \mathbf{1,540} \text{ billion euros} \quad (20)$$

(11) Equilibrium Consumption (C^*) (5 points)

$$C^* = 480 + 0.5(1,540) = 480 + 770 = \mathbf{1,250} \text{ billion euros} \quad (21)$$

Goods Market Clearing Verification:

$$Z^* = C^* + I + G = 1,250 + 110 + 250 = 1,610 \equiv Y^*$$

(12) Savings and Investment Accounting (5 points)

• Private Savings (S_{private}):

$$S_{\text{private}} = Y_D^* - C^* = 1,540 - 1,250 = \mathbf{290} \text{ billion euros} \quad (22)$$

Equivalently, via the private savings function:

$$S_{\text{private}} = -c_0 + (1 - c_1)Y_D = -480 + (1 - 0.5)(1,540) = -480 + 770 = 290 \quad (23)$$

• Public Savings (S_{public}):

$$S_{\text{public}} = T - G = 70 - 250 = \mathbf{-180} \text{ billion euros} \quad (\text{Budget Deficit of 180}) \quad (24)$$

• Investment Spending (I):

$$I = \mathbf{110} \text{ billion euros} \quad (25)$$

IS Identity Check: Total national savings equals investment:

$$S_{\text{total}} = S_{\text{private}} + S_{\text{public}} = 290 + (-180) = 110 \equiv I$$

(13) Government Spending Multiplier ($\Delta Y/\Delta G$) (5 points)

Holding taxes fixed ($\Delta T = 0$):

$$\frac{\Delta Y}{\Delta G} = \frac{1}{1 - c_1} = \frac{1}{1 - 0.5} = \frac{1}{0.5} = \mathbf{2} \quad (26)$$

A one-unit expansion in government spending increases equilibrium output by **2 units**.

(14) Tax Multiplier ($\Delta Y/\Delta T$) (5 points)

Holding government spending fixed ($\Delta G = 0$):

$$\frac{\Delta Y}{\Delta T} = \frac{-c_1}{1 - c_1} = \frac{-0.5}{1 - 0.5} = \frac{-0.5}{0.5} = -1 \quad (27)$$

A one-unit increase in lump-sum taxes contracts equilibrium output by **1 unit**.

(15) Balanced Budget Multiplier ($\Delta Y/\Delta G|_{\Delta G=\Delta T}$) (5 points)

Under a balanced budget fiscal expansion where $\Delta G = \Delta T$:

$$\left. \frac{\Delta Y}{\Delta G} \right|_{\Delta G=\Delta T} = \frac{\Delta Y}{\Delta G} + \frac{\Delta Y}{\Delta T} = \frac{1}{1 - c_1} + \frac{-c_1}{1 - c_1} = \frac{1 - c_1}{1 - c_1} = 1 \quad (28)$$

Output expands dollar-for-dollar with government purchases (Haavelmo's theorem), because government spending stimulates aggregate demand directly, whereas households absorb part of the tax hike by reducing private savings.

(16) Income-Dependent Taxes and Recalculated Multiplier (10 points)

Suppose taxes depend endogenously on aggregate income:

$$T = t_0 + t_1 Y = -252 + 0.2Y, \quad \text{where } t_0 = -252, t_1 = 0.2 \quad (29)$$

Substituting this tax rule into aggregate demand:

$$Y = c_0 + c_1[Y - (-252 + 0.2Y)] + I + G \quad (30)$$

$$Y = c_0 + c_1(1 - 0.2)Y - c_1(-252) + I + G \quad (31)$$

$$[1 - c_1(1 - t_1)]Y = [c_0 - c_1 t_0 + I + G] \implies Y = \frac{c_0 - c_1 t_0 + I + G}{1 - c_1(1 - t_1)} \quad (32)$$

Differentiating with respect to G :

$$\frac{\Delta Y}{\Delta G} = \frac{1}{1 - c_1(1 - t_1)} = \frac{1}{1 - 0.5(1 - 0.2)} = \frac{1}{1 - 0.4} = \frac{1}{0.6} = \frac{5}{3} \approx 1.67 \quad (33)$$

The government spending multiplier falls from 2.00 to **1.67** (or 5/3).

Remark on Baseline Equilibrium: At the original equilibrium $Y^* = 1,610$, endogenous tax collection equals $T = -252 + 0.2(1,610) = 70$. Thus, baseline equilibrium output is preserved at 1,610, but output responsiveness to exogenous shocks is substantially stabilized.

(17) Macroeconomic Rationale for Automatic Stabilizers (5 points)

An income-dependent tax system functions as an **automatic stabilizer** through two mechanisms:

1. **Built-in Cushioning of Disposable Income:** When an adverse shock reduces aggregate output ($\Delta Y < 0$), tax revenue automatically declines by $\Delta T = t_1 \Delta Y < 0$. Disposable income falls by only $\Delta Y_D = (1 - t_1)\Delta Y$, ensuring that $|\Delta Y_D| < |\Delta Y|$. Consequently, private consumption declines less severely than it would under lump-sum taxes.
2. **Damping of Multiplier Dynamics without Policy Lag:** The expenditure multiplier is lowered from $\frac{1}{1 - c_1} = 2.0$ to $\frac{1}{1 - c_1(1 - t_1)} \approx 1.67$. Because this damping operates endogenously and instantaneously without administrative, legislative, or implementation lags, it reduces business cycle volatility.